

Aptiv 2024 | CES 2024 Investor Event

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WELCOME MESSAGE

Kevin Clark | President and Chief Executive Officer

Hello, everybody. Welcome to Aptiv's pavilion here at CES 2024. I'll provide a little bit of background, then a few of my colleagues will dive deeper. We're going to take a brief break and allow you to get some refreshments and come back into the pavilion and spend time with our subject matter experts at each one of the stands here to talk about various aspects of our overall technology.

So for Aptiv, Delphi for those who've been around long enough, this is actually the 27th year that we've been here at CES, where we've presented content and material. CES is really an outstanding venue to get together with our customers, with our suppliers, with our partners to really talk about where the industry is headed, technology roadmaps, and where we align and discuss new business opportunities.

To put our week in context for all of you, this year we'll host over 500 customer events covering automotive, telecommunications, and other markets – actually a 20% increase over last year and several of those engagements are at very senior level / C suite levels within our automotive customers. And this year, it's with more than a dozen OEMs.

We're also hosting 35 vendors from across the globe, most from the semiconductor supply chain space, including a number of both established as well as emerging Chinese SOC providers that are providing current applications in the Chinese market, as well as developing solutions with us to bring those applications into the automotive market -- in the US and in Europe, in addition to China.

All of this is in addition to meetings with strategic partners, with regulatory agencies, with industry, analysts, media and others, so for us, it's a great opportunity to interact. There's a lot of activity. It's a very busy, busy week. Now since our launch of Aptiv in late 2017, we've focused on presenting fully functioning, productized offerings at CES. Solutions that our customers or supplier partners could actually see operate, could touch, could actually get in vehicles and enjoy the experience.

And this year is no different. So everything you see on display here today is really a part of our broader open architecture, both from a software and hardware standpoint, and really reflects our strategy to enable software-defined vehicles, which we talk a lot about -- from development all the way through operations.

Now, we all know there's a lot of noise regarding the automotive industry's challenges: executing the transition to electrified, feature-rich vehicles. We are really focused on how do we provide the cost effective solutions, which better enable the industry's transition, both from an operational standpoint as well as a cost standpoint, while providing them with increased flexibility. You'll see that today, we'll talk about that. All of this is on display here at CES, including vehicles running Aptiv's software features and Wind River embedded software on automotive grade hardware. They are actually being driven on the roads in Las Vegas with our customers.

So what you'll see at CES represents our portfolio that really allows for the creation, the development of full system solutions for our customers, from the sensor on the vehicle all the way to the cloud. And it's really a very comprehensive representation of how our technologies come together and enable the software-defined vehicle.

And a few examples are: Wind River's cloud-native on vehicle software platform which supports modularized, containerized applications that really leverage real-time safety certified solutions -- all of which are running on automotive-grade hardware, including all the high-speed, the high-fidelity signal and power and perception solutions that are really required for those technologies to run safely and effectively on cars.

Now, these solutions can be tailored to meet individual needs of a particular customer, as well as support our truly optimized platforms, including our Gen 6 ADAS solution where we're demonstrating hands-free, urban point-to-point navigation to customers here at CES. And I'd say there's really no substitute from a marketing and sales standpoint than getting actual technology in the hands of our customers. So everything you see here today, including the vehicles that we have out on the road, actually post-CES will be out at customer events in the US, in Europe, and in China, where they'll actually we have the opportunity to experience them firsthand.

So, now here to expand on my initial comments are a few my colleague colleagues, Benjamin Lyon, our Chief Technology Officer. He'll provide some context on how the intelligent edge is unlocking innovative new solutions that really, with the addition of Wind River, perfectly positioned Aptiv.

Paul Miller, Wind River's Chief Technology Officer. He's going to provide an example on how Wind River Studio and VxWorks are being used in the telecommunications market today, and how that really transitions and the optimal opportunity within the automotive space and what we're doing.

And then finally, Anant Thaker, our Chief Strategy Officer, our Senior Vice President of Product. We'll provide some background on what you see here in the technology theater and why it matters. And after that, we'll take a brief break, we'll allow you to get refreshments come back in at each one of these stations will have subject matter experts who will do a presentation and then be prepared to answer any questions. So with that, I'll turn it over to Benjamin.

THE SOFTWARE-DEFINED FUTURE

Benjamin Lyon | Senior Vice President and Chief Technology Officer

Thank you, Kevin! Our vision of the software-defined future is really taking hold! And we see that transformation which started with consumer products now increasingly impacting the broader IoT...

Unlocking this vision across multiple industries will require enhanced processing, intelligent software and a deep integration with the toolchains and connectivity platforms that support them. The devices around us are getting smarter and consuming more and more data. Additionally, consumers are demanding that products are continuously connected and always up to date. Just think about how we interact with our smart phones and the internet every single day!

The deployment of 5G is enabling more and more intelligence to move from the core cloud to the edge. This moves the processing of larger data volumes toward the source, facilitating quicker informed decisions and low latency cooperation between devices. To enable this, we must address the common technical challenges that lie ahead and demand continuous innovation... These are: enhancing processing power; upgrading software capabilities; and enabling true edge-to-cloud connectivity.

Today, the industries undergoing this digital transformation are coping with a shared set of challenges for both software and hardware: Complexity, Performance and Affordability. Now let's talk through these! Starting with Complexity: the legacy approach to increasing demand for new features and functions is more hardware... more boxes, each running their own embedded, non-standardized software.

And you can imagine how this struggles to scale as the feature list grows, and this leads to our next challenge, which is the impact this has on Performance. Integrating all of this heterogeneous hardware and software leads to lengthy software update cycles, inefficient use of processor resources, and challenges ensuring reliability.

There are literally multiple operating systems running on the vehicles we drive every day. The consumer product equivalent would be running Windows, Linux, and iOS all on your computer at the same time, and expecting it to work seamlessly. And the difference in Automotive and other mission-critical systems is that this needs to work every time and all the time. Fundamentally, all of this leads to challenges in affordability that effects both initial cost as well as the burden of ongoing updates.

To enable this transition, these challenges must be addressed with a modern, scalable architecture. In short, we need the right hardware, the right software, and the right toolchain. These three key puzzle pieces create a blueprint applicable across multiple industries and unlocks value creation and new business models: Starting with hardware, we need to simplify. This means up-integrating compute, designing processors for mixed-criticality workloads, and enabling software virtualization. In software, we must modularize to enable rapid updates and standard interfaces. We do that leveraging microservices and containers.

And lastly, all of this must be supported by a capable toolchain that enables scalable, software-on-device to software-in-the-cloud complete lifecycle management. We call this the edge-to-cloud connectivity platform. By adopting these solutions, our customers can not only navigate the transition to a software-defined future, but also lower cost and pave the way for future growth in their own software offerings.

As mentioned earlier, several industries are transitioning towards a software-defined future, albeit at different levels of maturity... Leading is consumer electronics. Products like your smart phone and edge-enabled features like video meetings are examples of mature deployments of cloud-native, edge enabled software-defined architectures. So we see in consumer products all three architectural puzzle pieces developed and deployed at scale.

Next is the telecommunications market, where this architecture is in initial production and scaling up. And it is followed by other industries, including aerospace and defense, automotive and industrial, with each at varying levels of maturity. Auto is clearly in the early stages of this evolution and OEMs continue to struggle to build, deploy, and keep complex software up to date. Aptiv and Wind River are perfectly positioned to help solve those key challenges.

And we're doing this by providing our customers with a comprehensive and flexible suite of solutions. In fact, we are the only full system solution provider in the automotive industry that can provide complete edge-to-cloud capabilities!

Let's start at the hardware foundation of the technology stack and work our way up... Aptiv offers industry-leading capabilities in advanced compute and electrification solutions we call our Smart Vehicle Architecture. This provides a flexible, scalable vehicle architecture that centralizes compute and optimizes power as well as data distribution. Importantly, this breaks the dependency of the software on the underlying silicon and allows greater flexibility in managing your silicon choices.

Moving up the stack to Cloud-Native Software – Wind River's on-vehicle solutions support microservices and containers for seamless software updates. This provides significant advantages for Aptiv's application software – allowing active safety, UX and other software to be continuously updated.

And lastly for Edge-to-cloud Connectivity – Wind River Studio is a suite of solutions that provides a DevOps toolchain that streamlines the entire software development lifecycle. This enables seamless and secure development, test, deployment and operation of complex software ecosystems.

We're really excited about the value we are bringing to this space, and to tell you more about it, I'd like to hand it over to Paul Miller, Wind River's Chief Technology Officer.

WIND RIVER DEMONSTRATOR

Paul Miller | Wind River's Chief Technology Officer

Thanks, Benjamin! I'm really excited to be here today. As we go to the first slide, you'll see a variety of the assets that Wind River brings to the table that are being used throughout the software-defined vehicle and software-defined verticals that Benjamin just took you on a tour of.

I'm going to start across the bottom you can see some of our infrastructure assets, VX Works, which is a safety-critical certified embedded operating system for real-time mission critical systems. Helix, our hypervisor that can be used to leverage virtualization to host virtual machines within these types of compute systems, as well as Wind River Linux, a Linux based on Yocto that we bring to the market to solve embedded Linux problems.

Now, why is this interesting? And why is it relevant to this vehicle that you see here? Wind River has been in business for 40 years now and have been engaged in a variety of different market verticals, as Kevin mentioned, aerospace and defense, automotive, industrial, medical, telecommunications. We find the software problems as a pure software company to be amazingly parallel between these different industries.

As we look at the aerospace sector over 10 years ago, one of our customers came to us and said, hey, I need to up-integrate the software applications that I have across a variety computers in my vehicle platform. I need to reduce the total cost of ownership of the system. I need to drop complexity of the physical assembly of the unit. I need to simplify the cable harnessing in my system to reduce weight and improve fuel economy.

The interesting thing was this company was Boeing. And so for over 10 years, our company has been involved in an integrated, fully virtualized solution that looks identical to the system that you see here, on the very planes that you flew on to come here today. With Boeing, with Airbus, with the customers that we're proud to have. We've been implementing this architecture for years. And so it's really interesting that we're able to use some of the assets that you see here to solve an identical problem in the automotive industry.

Additionally, we also know that, as you've seen, the new software-defined world is coming to bear. A lot of customers that we have, especially in the automotive space, are really challenged with software development. So some of the newer assets that we're bringing to the market, include Wind River Studio Developer that you see on the left hand side.

Later on after the presentation, you'll be able to see a demonstration run over here, where you'll actually watch Wind River Studio Developer walk through this left hand side diagram from requirements generation through coding, building that code, security scanning the code for cyber security risks, testing that code in a virtual test and physical testing environment, and then releasing that code as a software artifact that then we can deploy to a vehicle.

That same approach is used in industrial, medical, telecommunications, automotive – a tool chain that solves the developer paradigm. And as we know, from looking in the automotive industries, there are companies spending billions of dollars today, and failing to deliver on the promise of software-defined systems.

They not only need the assets that you see at the bottom layer here, to provide the right infrastructure environment to build the architecture of this software-defined vehicle. They also need a tool chain that enables them to actually build and deliver at high scale.

You may have 5,000 software developers across your development team to be able to deliver those software assets that are delivered as applications into that software-defined vehicle. So this is a really important offering that we hope you'll take a look at after the presentation.

On the right hand side, you can see after you have the software artifact and you release that software artifact, operating system, or application container, you then have to deploy it to something. In this case, we're deploying it to a vehicle, could be a medical device, could be a telecommunications 5G Edge infrastructure, doesn't really matter. It's a software asset being deployed to a device. So how do you do that? That's a different problem than developing a software that you see on the left. It's deployment and operations.

So on the right hand side, you see Wind River Studio Operator, it's a set of software assets that lets you deploy and operate software at a very, very high scale. And this has really contemporary software capabilities that allow us to deploy manage software across millions of devices.

You also see after deployment, we enter operate and monitor. The operations really refers to the continuous updates that you apply to these remote systems, cybersecurity updates, and new functionality that can generate new revenue for our customers. These are enabled through over-the-air updates and the ability to connect to these remote devices, and manage software on those systems throughout the entire lifecycle of that system.

And then finally monitoring – the ability to take a digital feedback loop to collect data from those edge systems. You'll see some of that today on the demonstration to my left here, where high performance aspects of our embedded software systems also provide data live feedback from these edge devices. These can inform digital twin systems to allow you to virtually rebuild systems and re-release software to those edge devices, as well as collect important data on the operation of systems as they're deployed.

Now, what you're seeing here, in this picture, is actually a well-known software concept. It's called DevOps or DevSecOps, if you include security. If you look into modern software development environment, this is a normal thing that is the current way that software is being built. It really embraces the entire challenge that you have for software development, from building the code from requirements, releasing it, deploying it to your vise and operating it and then closing that loop to iterate that code and continuously deployed to the system.

These are the principles and assets that other industries have adopted. And it's critical for the automotive industry, as they become more and more complex with software-defined architectures to truly deliver on the promise of what the system can do. *Next slide.*

As we take a look at another example, in telecommunications, some of you may be aware that Wind River is actively involved in next generation 5G deployment. Some of our infrastructure assets based on Linux and Kubernetes are being used by Verizon to deploy their entire 5G network across the entire domestic United States. We also work with Vodafone in the UK, and KDDI and NTT Docomo. They had the exact same problem, just from a telecommunications perspective.

I need to deploy assets in a public cloud environment that are doing back office billing infrastructure and management of these deployed systems in their enterprise and their regional data center, in what they call a NOC or network-operations-center, they're deploying software assets that run their infrastructure.

And then finally, at the cell tower, they're deploying a virtual distributed unit. That's a virtual radio access network (vRAN) that allows them to move from a proprietary bare metal system that was an appliance to now a fully cloud-based Kubernetes solution for deploying the entire 5G infrastructure. And Wind River's deploying that at scale today. We're the only company doing that today.

You'll notice a very similar problem here, I've got 10s of 1,000s of cell towers, maybe even hundreds of 1,000s. I've got to deploy an update software to that edge system. That's critical to enable high availability for services like emergency 911 calling. And many of you standing here that have Verizon cell phones are actually connected today into the Wind River Aptiv solution is powering that network.

You'll see on the right hand side, as I've been speaking, the diagrams that you're seeing there are showing the real-time state of the deployment of the software from a single pane of glass that's remote, deploying software into a public cloud, private cloud and edge devices -- all from a single pane of glass, from a single user interface. This really solves the operational need. If you look at the total cost of ownership for deploying and managing a software system, your upfront costs are important in the initial development. But the cost and the opex is dominated by the long term support and management of that software. So you really need to be able to deploy and operate software, as well as design and build it. *Next slide.*

If we take a deeper look, you'll find other parallels here. This is a vertical stack that shows how the system is architected, and compares the telecommunication solution on the left to the software-defined vehicle architecture on the right hand side.

As you see on the left, you start at the bottom with optimized hardware. That telecommunications infrastructure that I just referred to, as built on commercial off-the-shelf Intel powered systems from the likes of Dell and Hewlett Packard. Those are server architecture systems that are deployed using this cloud infrastructure at the far edge of the network. On top of that system, you then see Wind River's infrastructure assets being deployed across that, to host the applications that actually provide the 5G functionality.

So Wind River kind of sits in the middle of a layer cake where at the bottom you have hardware, then you have infrastructure, and then on top of that you have the radio access network, from companies such as Samsung, Ericsson and Nokia. Now those names may be familiar to you, because you've heard of those type of companies providing telecommunications infrastructure. They used to provide it as a bespoke appliance, a hardware-based appliance.

As we now move towards cloud-native microservices and containers, it's being developed and deployed as a completely cloud-native architecture running on conventional server software, with our applications in there running and providing the Kubernetes and Linux infrastructure that hosted those applications. Again, a very similar thing to what we talked about in the automobile.

And then at the top, you see Wind River Studio, the dashboard that we were just looking at providing the ability to remotely manage and continuously update that software that's in that deployed architecture. As we move to the right hand side, and you look at the vehicle in front of you, it's a very similar problem. You've got general purpose compute, it's up integrating and virtualizing applications, in this case provided by Aptiv.

Using Wind River technology, again, in the same way that's used in the telecommunications or in the aerospace example I gave previously, to virtualize these environments and now up integrate in a vehicle that used to have 150+ different ECUs/computers to one that now has less than 10. That's the exact same architecture that we talked about from the avionics industry and aerospace industry. Then on top of that Wind River Studio, providing the operational ability to provide over-the-air updates and manage those systems.

You'll see in the demonstration here later today, not only Studio developer capability with building the software, but actually using over-the-air update into this vehicle to dynamically deploy a container with that updated software using these operational capabilities. And so Studio provides that development environment, as well as the operational environment delivered with a set of assets that enable you to build these architectures for many different industries. Next slide.

Interestingly, as well, you can see this paradigm of core to edge or sensor to cloud, where on the left hand side, you can see public cloud, the right hand side you see the device. And again, that device could be an automobile, a medical device, an industrial robotic system or a telecommunications infrastructure. The importance of this diagram comes from the fact that as people think about building software systems, they're no longer thinking about just the device. We tend to get raptured by looking

at the vehicle and thinking about what is going on in the vehicle. But I can promise you the software that's being developed here is connected to backend cloud systems.

So when we look at Studio developer and the ability to build the software applications, we're deploying components of that in public cloud, in enterprise, at far edge of telecommunications networks, and in the actual vehicle itself. And managing that as an end-to-end system is an incredibly important thing.

Next slide.

And finally, as we move into the future, what's coming after this? If this technology is proven, what's going to happen as we bring this forward? A couple of things that are interesting... we talked about the use of our technology in the 5G infrastructure that's being deployed globally, and the use of our technology in-vehicle. Now these things can be brought together by one company, the first company in history that's ever had these capabilities together to bring things like cellular V2X or cellular vehicle-to-everything.

This is the software paradigm where I can bring infrastructure intelligence from deployed infrastructure in civil environments, traffic light status, intersections, construction status, intelligently into vehicle to affect its routing decisions and self-driving capabilities. I can do vehicle-to-vehicle accident avoidance, and other applications that are very real-time between vehicles that are dynamically moving in traffic by connecting directly from the 5G cell tower.

So as you move toward these more cloud-native capabilities and containerized architectures, as you can manage the software across a distributed environment, you can now establish connectivity between that 5G edge infrastructure and the vehicles that are traversing around connected to it.

Similarly, in the center diagram, the concept of offloading compute, now that the vehicle has a containerized application, that's the same container - an OCI compliant container - that's running in the vehicle that could be running in Amazon EKS, or in an edge of a telecommunications infrastructure. So I can move that application around. Even though to the user the vehicle, that in vehicle infotainment looks like it's in the car, but it might be running on a local cell tower or in a centralized cloud application. The software can be dynamically moved.

This is important for a vehicle later on in its life. You may want to bring new applications to it, but there's not enough compute in it. By leveraging edge cloud compute, you can now scale the functionality of that environment and build new revenue models as an automotive OEM. Whereas before you were locked into the compute that was in the vehicle.

And then finally, on the right hand side, private 5G networks also very exciting, especially for the industrial sector, manufacturing environments. Having reliable connectivity that is superior to WiFi, or 4G LTE, with quality of service with high endpoint count capability, allows you to do cloud robotics control within the automotive manufacturing sector. It also enables you to do things like valet parking within parking lots of these connected and self-autonomous vehicles. So some really interesting applications here as we go forward.

As we start with the foundation of this technology and a software-defined vehicle, we'll bring that forward into next generation applications that bring together what we're doing in telecommunications and aerospace to build the ultimate solution here for a software-defined system. Thank you, everyone.

WHAT YOU'LL SEE TODAY: APTIV TECHNOLOGY THEATER

Anant Thaker | Aptiv's Chief Strategy Officer and Senior Vice President of Product

Thanks, Paul, and hello everyone! So Benjamin described how digital transformations across multiple industries have the same core needs. And these needs can only be addressed with an edge-to-cloud DevOps platform, cloud-native software on these distributed mission-critical devices, and optimized hardware architectures that enable that software-defined functionality. Paul then showed us how Wind River is already addressing these needs with its Studio and Edge offerings in industries like telecommunications.

As you all know – the automotive industry is struggling with software: it's taking too long to develop, it's too expensive, it has too many bugs, and it can be significantly impacted by disruptions in the semiconductor supply chain. So we're excited to showcase – both inside the pavilion and in the vehicles outside – how we are directly addressing these challenges, and how Aptiv and Wind River are uniquely positioned to provide our customers with all of the products and technologies they need to truly enable software-defined vehicles.

Now you'll be able to explore all of our latest innovations and engage directly with our subject matter experts, but first let me highlight some of the things you'll be seeing. I'll start at the top of the stack with our first station here on my right...

In our first technology display, you'll see how we're bringing Wind River Studio to automotive applications. We have a live demo in which Studio is connected to our vehicle skateboard here, and we'll show you how it enables Aptiv, our customers, and our partners to develop software faster and at lower cost, to seamlessly and securely deploy that software to vehicles, and to derive real-time insights for continuous software lifecycle management.

You'll see that Studio is an open, modular, and flexible platform – today it has more than 30 commonly used third-party tools pre-integrated and available through Studio Gallery, with many more on the way. And customers can also bring their existing toolchains into the Studio environment. This enables software developers – who are often in working in distributed teams across the world – to code, build, scan, and test their software applications using the tools they already know– but do so much more efficiently with optimized pipelines . And when you add in cloud-scale automation and easy access to virtual test environments –Studio can significantly increase developer productivity and enable bugs to be detected and fixed much earlier in the development cycle.

Now importantly, you'll see that all these end-to-end functionalities are offered a single pane of glass. And we're confident these solutions can reduce the costs of our OEM customers in software development, deployment, and post-production lifecycle management by more 25%... As well as support their new business models by enabling more frequent software releases for new features and performance enhancements.

But enabling software-defined vehicles requires not just a modern toolchain – you also need a modern software architecture and platform in the vehicle itself. And that's what we showcase in the second technology display.

This is the software that sits between the underlying compute and silicon below and the application software above. It's needed to abstract from and virtualize a vehicle's underlying compute hardware, to host advanced software applications, including mixed-criticality workloads, and to enable features

updates entirely through software – all while guaranteeing real-time behavior for safety critical applications like ADAS.

Wind River's Edge software platform includes the only real-time operating system to support services-based software architectures and OCI containers at the highest levels of safety certification. This means customers can deploy containerized applications directly to the vehicle and improve software updateability, portability, and re-use. And they can do this all without any sacrifices to functional safety.

Importantly – in this display, we have Aptiv's ADAS and user experience software running on the Wind River Edge platform across multiple semiconductor chips, and we show how its performing on those chips using Studio-enabled analytics. This provides our customers with significant flexibility and choice in SoCs, which are one of the most expensive components in their supply chain and where resiliency is paramount

Then at our next station, you'll see how Aptiv is deploying containerized software building blocks and applications onto the Wind River Edge platform. Aptiv continues to be a leader in both exterior and interior perception. We were the first to put a radar on a vehicle, the first to do sensor fusion, and the first to apply AI/ML to in-cabin sensing. Here, we are showcasing how, by taking a system-level approach to ADAS that optimizes multiple sensing modalities, we can deliver high performance perception with efficient compute requirements that therefore enable lower costs.

For example, by applying machine learning to enhance our radar perception performance and leveraging our advanced fusion capabilities, we can provide highly robust perception for L2+ ADAS applications that requires 65% lower compute and power consumption than a camera-dominant configuration. This is a core differentiator of our Gen 6 ADAS platform, which enables us to deliver the solution at 25% lower cost than competitors.

And because this software stack is hosted on the Wind River Edge platform, we can deploy targeted, containerized updates to modules of our Gen 6 ADAS stack – such as regular ML-driven improvements to radar perception – all with the highest levels of safety certification.

So we've talked about our DevOps platform and in-vehicle software stack – both of these are further enabled by our optimized, next-generation hardware architecture highlighted right here in front of me and the high voltage station to my right. Our high voltage display showcases our comprehensive portfolio of high voltage power distribution, power conversion, battery management, and connection systems

Specifically you'll see our industry-leading busbar technology, which enables more efficient power distribution and optimized battery pack design and manufacturing, our integrated power electronics solutions, which up-integrate the on board charger, BDU, and DC-DC converter; and our innovative, containerized battery management software, which enables safe, fast charging while minimizing battery degradation. The common factor across all of these solutions is enabling electrification while reducing complexity, weight, and mass requirements and lowering total system cost.

Bringing this all together right here in front of me is our foundational Smart Vehicle Architecture – Aptiv's sustainable, open platform that enables feature-rich, software-defined, electrified vehicles while providing 25% savings in weight and packaging space for compute, 75% lower system integration & testing cost, and 50% reduction in labor for electrical architecture installation.

We have over \$10B of new SVA business bookings to-date and expect nearly 25% of all vehicles produced in 2030 to have “SVA-like” architectures. And this architecture – both hardware and software – also serves as a foundation to connect vehicles into the broader intelligent edge, which Benjamin and Paul talked about earlier.

Some of you may remember the first time we discussed our SVA approach in 2017 and the skateboard we unveiled at CES in 2020... But it has never previously been used to power a vehicle drivable on public roads and supporting safety-critical applications such as ADAS - That is, until now... Outside, for our customers, we are demonstrating.

First, our Software-Defined Vehicle, which features our full, containerized Gen 6 ADAS software stack, running on the Wind River software platform – all on the foundation of our Smart Vehicle Architecture. I want to emphasize that this is not a PC-based demonstration with the equipment filling up the trunk. This is a fully embedded, automotive grade system showcasing the full scope of our software stack on public roads.

And second, our high voltage demonstrator, which includes the full breadth of our signal and power solutions portfolio, including high and low voltage architecture, our battery management software, and solutions from Intercable Automotive Solutions and HellermannTyton – all fully integrated into functional 800V vehicle architecture.

So to wrap up: As multiple industries transition to their software-defined future, we see the same core solutions enabling it: an edge to cloud DevOps platform, cloud-native software stack, and optimized hardware architectures. Wind River has already been at the forefront of bringing these solutions to multiple industries.

The automotive industry has struggled with the software transition, and it needs this modern architecture and development platform to truly shift to software-defined vehicles. And Aptiv and Wind River are uniquely positioned to enable this, as we hope the technologies we are demonstrating today proves.

So with that, I'd like to invite you to get a refreshment in the Aptiv café. We're going to move the chairs out of here to open up the space, and we invite you to explore all of our innovations in greater detail. Thank you.

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